```
<!-- Dynamic field definitions allow using convention over configuration
   for fields via the specification of patterns to match field names.
   EXAMPLE: name="*_i" will match any field ending in _i (like myid_i, z_i)
   RESTRICTION: the glob-like pattern in the name attribute must have
   a "*" only at the start or the end.  -->

<dynamicField name="*_i"  type="int"    indexed="true"  stored="true"/>
<dynamicField name="*_is" type="int"    indexed="true"  stored="true"  multiValued="true"/>
<dynamicField name="*_s"  type="string"  indexed="true"  stored="true" />
<dynamicField name="*_ss" type="string"  indexed="true"  stored="true"  multiValued="true"/>
<dynamicField name="*_l"  type="long"   indexed="true"  stored="true"/>
<dynamicField name="*_ls" type="long"   indexed="true"  stored="true"  multiValued="true"/>
<dynamicField name="*_t"  type="text_general"    indexed="true"  stored="true"/>
<dynamicField name="*_txt" type="text_general"   indexed="true"  stored="true" multiValued="true"/>
<dynamicField name="*_en"  type="text_en"    indexed="true"  stored="true" multiValued="true"/>
<dynamicField name="*_b"  type="boolean" indexed="true" stored="true"/>
<dynamicField name="*_bs" type="boolean" indexed="true" stored="true"  multiValued="true"/>
<dynamicField name="*_f"  type="float"  indexed="true"  stored="true"/>
<dynamicField name="*_fs" type="float"  indexed="true"  stored="true"  multiValued="true"/>
<dynamicField name="*_d"  type="double" indexed="true"  stored="true"/>
<dynamicField name="*_ds" type="double" indexed="true"  stored="true"  multiValued="true"/>

<!-- Type used to index the lat and lon components for the "location" FieldType -->
<dynamicField name="*_coordinate"  type="tdouble" indexed="true"  stored="false" />

<dynamicField name="*_dt"  type="date"    indexed="true"  stored="true"/>
<dynamicField name="*_dts" type="date"    indexed="true"  stored="true" multiValued="true"/>
<dynamicField name="*_p"  type="location" indexed="true" stored="true"/>

<!-- some trie-coded dynamic fields for faster range queries -->
<dynamicField name="*_ti" type="tint"    indexed="true"  stored="true"/>
<dynamicField name="*_tl" type="tlong"   indexed="true"  stored="true"/>
<dynamicField name="*_tf" type="tfloat"  indexed="true"  stored="true"/>
<dynamicField name="*_td" type="tdouble" indexed="true"  stored="true"/>
```

Figure 4: Dynamic fields

```
<!-- copyField commands copy one field to another at the time a document
     is added to the index.  It's used either to index the same field differently,
     or to add multiple fields to the same field for easier/faster searching.  -->

<copyField source="cat" dest="text"/>
<copyField source="name" dest="text"/>
<copyField source="manu" dest="text"/>
<copyField source="features" dest="text"/>
<copyField source="includes" dest="text"/>
<copyField source="manu" dest="manu_exact"/>

<!-- Copy the price into a currency enabled field (default USD) -->
<copyField source="price" dest="price_c"/>

<!-- Text fields from SolrCell to search by default in our catch-all field -->
<copyField source="title" dest="text"/>
<copyField source="author" dest="text"/>
<copyField source="description" dest="text"/>
<copyField source="keywords" dest="text"/>
<copyField source="content" dest="text"/>
<copyField source="content_type" dest="text"/>
<copyField source="resourcename" dest="text"/>
<copyField source="url" dest="text"/>

<!-- Create a string version of author for faceting -->
<copyField source="author" dest="author_s"/>

<!-- Above, multiple source fields are copied to the [text] field.
  Another way to map multiple source fields to the same
  destination field is to use the dynamic field syntax.
  copyField also supports a maxChars to copy setting.  -->
```

Figure 5: Copy fields

Figure 6: Query

is false; if true, it indicates that a single document might contain multiple values. 'required' default value is false—it instructs Solr to reject any attempt to add a document that does not have a value for this field. Figure 2 shows how to define field type.

Dynamic fields: Solr has strong data typing for fields, yet it provides flexibility using 'Dynamic fields'. Using the *<dynamic field>* tag declaration, you can create field rules that Solr will use to understand what data type should be used whenever it is given a field name that is not explicitly defined, but matches a prefix or suffix used in the dynamic field.

For example, dynamic field affirmation tells Solr that whenever it sees a name ending in '_i' which is not an explicitly defined field, then it should dynamically create an integer field like what's shown below:

```
<dynamicField name ="*_i"
type="integer" indexed="true"
stored="true"/>
```

Copy fields: The *<copyField>* declaration is used to instruct Solr that you want to duplicate any data it sees in the 'source' field of documents that are added to the index, in the 'destination' field of that document. But remember that datatypes of the fields are compatible. Duplication of fields is done before any analysers are invoked. By the use of copy field, Solr will be able to search by simply giving the value and not defining the tag. For example, if there is a field like *Roll number,* without the use of *copyField,* we can search by giving the command 'rollno:10'.

With the use of *copyField*, the syntax of the copy field declaration is:

```
<copyField source="rollno"
dest="text"/>
```

Now we can search just by typing '10' and Solr will give you the same result.